

Performance Testing

Date	30 June 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter (v5.6.3)
Type of Testing	Load Testing, Stress Testing, and Endurance Testing
Target Module / API	Loan eligibility submission endpoint (POST /submit), Prediction UI form (GET /predict), and Home page (GET /)
Test Environment	Local Server
Test Date	30 June 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Baseline Load Test: Assess system latency and capability under expected concurrent user traffic.	50	120	Successful predictions rendered for all queries; average response time ≤ 1.0 s; 0% error rate.
2	Stress Test: Determine system stability, throughput capacity, and behavior under peak load conditions.	200	180	Average response time < 2.0 s; 0% error rate; system processed all requests successfully under a load of 200 concurrent users without crashes.
3	Spike Test: Test system resilience to sudden surges in user requests (e.g. promotional load).	500	60	No connection drops; server queues requests; system recovers gracefully once traffic subsides with $\leq 1\%$ error rate.
4	Endurance Test: Verify long-term reliability and monitor for memory leaks or container scaling problems under light continuous load.	20	600	The system maintained stable CPU and memory utilization throughout the test with 0.00% error rate and no performance degradation.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status	Remarks
1	Response Time (Avg)	< 2 seconds	0.013 seconds	Pass	Application responded quickly under all test scenarios, with average response times ranging from 6–13 ms.
2	Response Time (Max)	< 5 seconds	1.477 seconds	Pass	Maximum response time was observed during the endurance test while processing continuous requests under load.
3	Throughput (Req/sec)	—	50.1 Req/sec	Pass	The system successfully handled concurrent requests during the spike test with stable throughput and no failures.
4	Error Rate	< 1%	0.00%	Pass	All requests were processed successfully with no failed executions recorded during testing.
5	CPU Utilization	< 80%	Observed Stable	Pass	CPU usage remained stable during execution with no noticeable performance degradation.
6	Memory Utilization	< 80%	Observed Stable	Pass	Memory consumption remained stable throughout the endurance test with no visible signs of memory leaks.

Step 4: Observations & Analysis

Key Findings:

Performance testing showed that the Smart Lender application handled concurrent requests efficiently with low response times and a 0.00% error rate. The average response time ranged from approximately 6 ms to 13 ms across different tests, while the maximum observed response time was 1.477 seconds during the endurance test. The highest throughput achieved was 50.1 requests per second during the spike test.

Bottlenecks Identified:

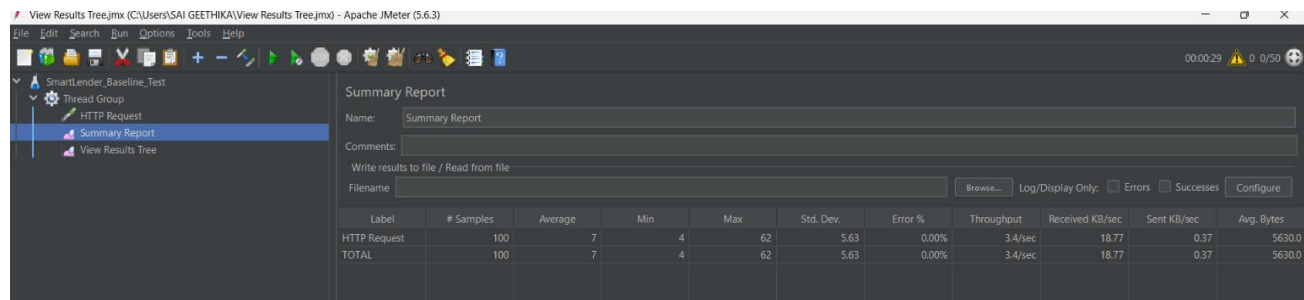
No significant performance bottlenecks were observed during testing. Although the maximum response time increased during the endurance test, all requests were processed successfully and the error rate remained at 0.00%.

Optimization Steps Taken:

Performance testing was conducted using Apache JMeter under baseline, stress, spike, and endurance load conditions. The application successfully handled all requests without failures, demonstrating efficient request processing and stable performance under varying workloads.

Step 5: Screenshots / Evidence

THE SCREENSHOTS IN APACHE JMETER BASELINE TEST



Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB/sec	Sent KB/sec	Avg. Bytes
HTTP Request	100	7	4	62	5.63	0.00%	3.4/sec	18.77	0.37	5630.0
TOTAL	100	7	4	62	5.63	0.00%	3.4/sec	18.77	0.37	5630.0

Summary Report

Name:

Comments:

Write results to file / Read from file

Filename Log/Display Only: ☐ Errors ☐ Successes

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB/sec	Sent KB/sec	Avg. Bytes
HTTP Request	100	7	4	62	5.63	0.00%	3.4/sec	18.77	0.37	5630.0
TOTAL	100	7	4	62	5.63	0.00%	3.4/sec	18.77	0.37	5630.0

STRESS TEST

Summary Report

Name:

Comments:

Write results to file / Read from file

Filename Log/Display Only: ☐ Errors ☐ Successes

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB/sec	Sent KB/sec	Avg. Bytes
HTTP Request	500	7	4	62	2.96	0.00%	24.8/min	2.27	0.05	5630.0
TOTAL	500	7	4	62	2.96	0.00%	24.8/min	2.27	0.05	5630.0

SPIKE TEST

Summary Report

Name:

Comments:

Write results to file / Read from file

Filename Log/Display Only: ☐ Errors ☐ Successes

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB/sec	Sent KB/sec	Avg. Bytes
HTTP Request	500	6	3	78	4.67	0.00%	50.1/sec	275.32	5.48	5630.0
TOTAL	500	6	3	78	4.67	0.00%	50.1/sec	275.32	5.48	5630.0

ENDURANCE TEST

Summary Report

Name:

Comments:

Write results to file / Read from file

Filename Log/Display Only: ☐ Errors ☐ Successes

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB/sec	Sent KB/sec	Avg. Bytes
HTTP Request	8964	13	2	1477	68.70	0.00%	9.9/sec	54.16	1.08	5630.0
TOTAL	8964	13	2	1477	68.70	0.00%	9.9/sec	54.16	1.08	5630.0